

CO2 AT1 - Analytical problem solving

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course: operating system
course code: CSA0401
Date: 30.7.26

1) CPU scheduling - waiting time & Turnaround time

A set of process arrive time 0 with following burst time

Process	Burst Time (ms)
P ₁	10
P ₂	4
P ₃	6
P ₄	2

20

- calculate waiting time & Turnaround time for each process using SJF (non preemptive)
- compute the average waiting time & average turnaround time
- Interpret whether SJF improve CPU utilization compared to FCFS.

2. Round-Robin

Process	Burst time (ms)
P_1	20
P_2	5
P_3	13
P_4	8

Time quantum = 4 ms

- construct the Gantt chart for Round Robin
- calculate waiting time & Turnaround time of each process
- Analyze how changing the quantum to 10 ms would affect system performance (CPU utilization & context switching)

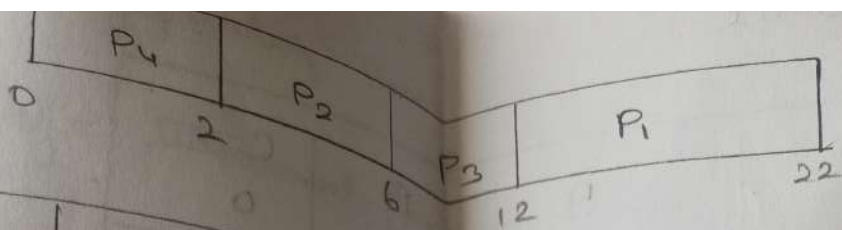
Deadlock - safety check using Banker's Algorithm.

A system has 3 resource types (A, B, C). Total 20

A = 10, B = 5, C = 7.

Total - Allocated

WOT = T₀ - B₀



22
6
12
2

Process	Burst Time	Turnaround Time CT - AT	waiting Time TA - BT
P ₁	10	22 - 0 = 22	22 - 10 = 12
P ₂	4	6 - 0 = 6	6 - 4 = 2
P ₃	6	12 - 0 = 12	12 - 6 = 6
P ₄	2	2 - 0 = 2	2 - 2 = 0

a)
10/16/2022

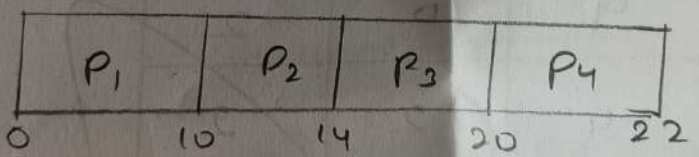
b)

Average waiting time $(12 + 2 + 6 + 0) / 4 = 20 / 4 = 5$

ATA $= (22 + 6 + 12 + 2) / 4 = 42 / 4 = 10.5$

c)

FCFS Gantt chart



Process	Burst Time	Turnaround Time	waiting Time
P ₁	10	10 - 0 = 10	10 - 10 = 0
P ₂	4	14 - 0 = 14	14 - 4 = 10
P ₃	6	20 - 0 = 20	20 - 6 = 14
P ₄	2	22 - 0 = 22	22 - 2 = 20

Process	A	B	C
P ₁	0	1	0
P ₂	2	0	0
P ₃	3	0	2
P ₄	2	1	1
P ₅	0	0	2

Maximum matrix : 5 2 5

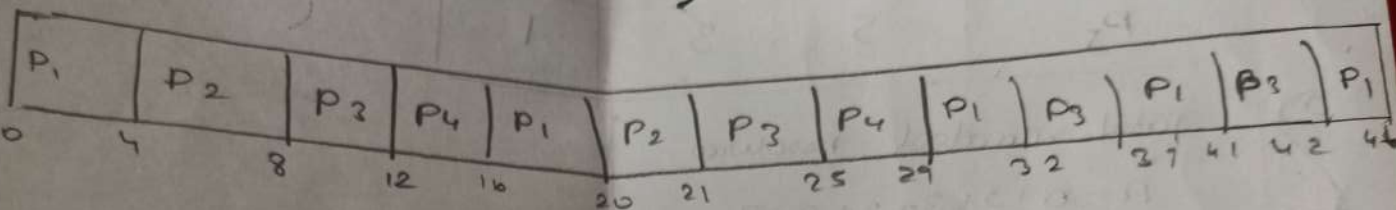
Process	A	B	C
P ₁	7	5	3
P ₂	3	2	2
P ₃	9	0	2
P ₄	4	2	2
P ₅	5	3	3

- calculate the Need matrix
- Determine the system in a safe state using using Banker's Algorithm.
- Provide a valid safe sequence if it exists.

* Average waiting Time = $(0 + 10 + 14 + 26) / 4 = 64 / 4 = 16$

* Average Turnaround time $(10 + 14 + 20 + 22) / 4 = 66 / 4 = 16.5$

* yes SJF improves CPU utilization compared to FCFS.



a)

Process	Burst Time	Turnaround Time	waiting time
P ₁	20	46	26
P ₂	5	21	16
P ₃	13	42	29
P ₄	8	29	21

Aug waiting Time $(26 + 16 + 29 + 21) / 4 = 92 / 4 = 23$

Aug Turnaround Time $(46 + 21 + 42 + 29) / 4 = 138 / 4 = 34.5$

$$\text{Need} = \text{Maximum} - \text{Allocation}$$

Process	A	B	C
P ₁	7	4	3
P ₂	1	2	2
P ₃	6	0	0
P ₄	2	1	1
P ₅	5	3	1

Total Allocated Resource.

$$A = 0 + 2 + 3 + 2 + 0 = 7$$

$$B = 1 + 0 + 0 + 1 + 0 = 2$$

$$C = 0 + 0 + 2 + 1 + 2 = 5$$

Available = Total - Allocated

$$A = 10 - 7 = 3$$

$$B = 5 - 2 = 3$$

$$C = 7 - 5 = 2$$

$\{3, 3, 2\}$

Available = $(3, 3, 2)$ + Allocation

$$= (3, 3, 2) + (2, 0, 0)$$

$$= (5, 3, 2)$$

Given :

Probability of waiting for I/O (P)

$$= 80\% = 0.8$$

$$\text{CPU time} = 20\%$$

$$\text{Formula } U = 1 - P^n$$

$$P = 0.8$$

n = Degree of multiprogramming

a)

$$U = 1 - P^n$$

$$U = 1 - (0.8)^4$$

$$[(0.8)^4 = 0.4096]$$

$$U = 1 - 0.4096$$

$$U = 0.5904$$

$$U = 59.04\%$$

b) Degree of multiprogramming = 6

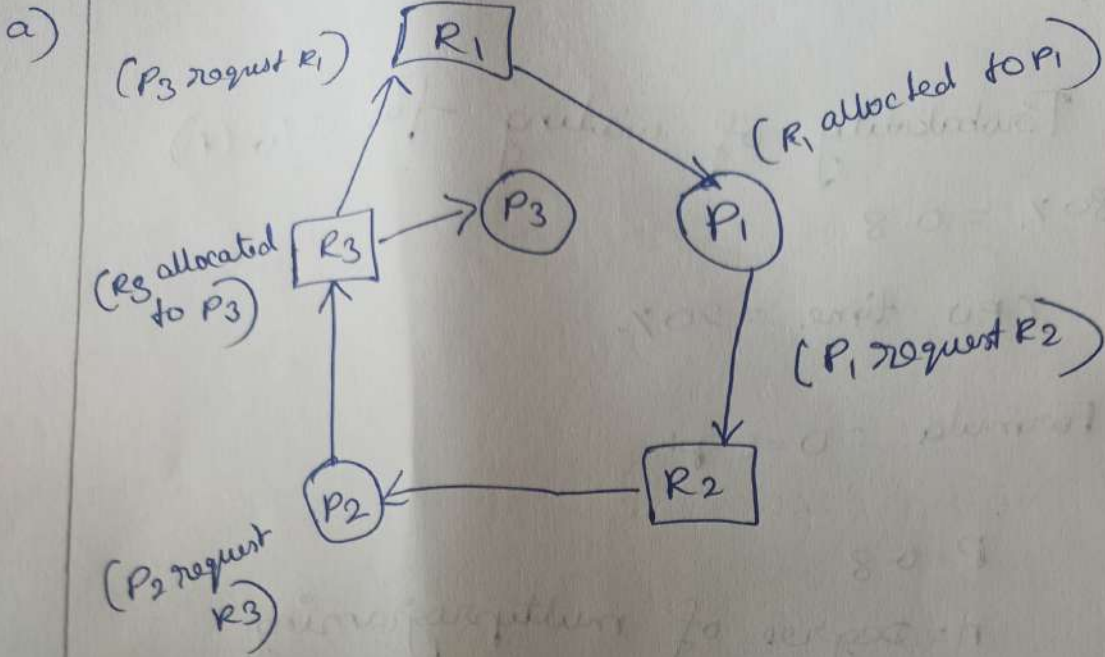
$$U = 1 - (0.8)^6$$

$$U = 1 - 0.262144$$

$$U = 0.737856$$

$$U = 73.79\%$$

4) (RA6)



b) $P_1 \rightarrow R_2 \rightarrow P_2 \rightarrow R_3 \rightarrow P_3 \rightarrow R_1 \rightarrow P_1$

each resource has one instance and it forms a cycle the system is deadlock.

c) Recovery method:

Terminate one of the deadlock process

After terminating P3:

R3 become free

P2 gets R3 & finish

R2 become free

therefore the deadlock is removed.